Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Main Topic: [Banana] – [Pest Management]

1.1 Banana Pseudostem Borer / Pseudostem Weevil (*Odoiporus longicollis*)

Common and Scientific Names, Taxonomy, and Classification

- **Common Names:** Pseudostem Borer, Pseudostem Weevil, Banana Stem Weevil.
- **Scientific Name:** *Odoiporus longicollis* (Olivier) / *Odoiporus longicollis* (Oliver).
- **Taxonomic Classification:** Order Coleoptera, Family Curculionidae.

Life Cycle and Developmental Stages

- **Egg:** Cream-colored, cylindrical in shape with rounded ends. The incubation period ranges from 3 to 8 days.
- **Larva (Grub):** Emerging grubs are fleshy, yellowish-white, and apodous (legless). Larvae bore into and feed on the succulent sheath tissues of the pseudostem, creating extensive longitudinal tunnels that can reach the central core.
- **Pupa and Adult:** Adults are stout weevils that are either reddish-brown or black in color.

Seasonal Occurrence, Favorable Conditions, and Host Range

- **Seasonal Activity:** The pest is active during warm summer and monsoon months.
- **Crop Vulnerability Stage:** Infestation typically initiates in 5-month-old plants and becomes most damaging during the flowering and bunch formation stages.
- **Varietal Susceptibility:** Stem weevil attack is particularly prevalent in cultivars such as Nendran, Red Banana, Karpuravalli, and Monthan.

Feeding Behavior, Symptoms, and Damage

- **Early Diagnostic Symptoms:** Presence of small, pinhead-sized oviposition/feeding holes on the pseudostem; fibrous extrusions from the bases of leaf petioles; presence of adult weevils; and exudation of a gummy, jelly-like, or slimy plant sap from stem holes.
- **Direct Larval Damage:** Grubs bore extensively through the pseudostem sheaths and bunch stalks, forming long, discolored larval tunnels. As larval feeding hollows out the pseudostem, affected plants suffer severe growth stunting, impaired bunch development, and pseudostem snapping/breaking at the apical region during strong winds. If larval tunneling passes through the central growing point, the plant is killed.

- **Secondary Rotting and Odor:** Primary larval tunneling encourages secondary invasion by saprophytic and rot-causing micro-organisms, leading to tissue decay and emission of a foul odor.

Field Monitoring

- Monitor adult stem weevil activity using longitudinal split pseudostem traps (20–60 cm or 30 cm long, placed with cut surfaces facing the ground) or disc-on-stump traps at a density of 10–15 traps/ha (or up to 40 traps/acre).
- Adult weevils are attracted to traps by host volatiles released from damaged pseudostems as well as male aggregation pheromones.

Integrated Pest Management (IPM)

1. Cultural and Mechanical Control

- **Field Sanitation:** Maintain clean cultivation by regularly removing and burning dried leaves, dead foliage, and surplus side suckers (prune side suckers monthly).
- **Eradication of Infested Plants:** Uproot and burn severely infested plants and corms. Do not dump infested plant residues into manure pits; cut plants at the ground level and treat the stump.
- **Post-Harvest Management:** Avoid *matocking* (leaving harvested pseudostems standing) in weevil-endemic areas. Cut harvested pseudostems into 30 cm lengths to serve as monitoring/collection traps rather than piling them in heaps. Treat harvested cut bases with 100 mL carbaryl (2 g/L).
- **Oviposition Deterrent Slurry:** In hotspot areas, apply a mud slurry mixed with 5% neem oil onto the pseudostem 5 months after planting to deter female egg-laying.

2. Biological Control

- **Biopesticide-Swabbed Traps:** Swab the cut surfaces of split stem traps with entomopathogenic fungi or nematodes and lay them cut-side down near plant bases:
  - *Beauveria bassiana* (20 g rice chaffy grain formulation, 3 mL in 100 mL liquid formulation, or 1×10^8 spores/mg), or *Metarhizium anisopliae*.
  - Entomopathogenic nematodes (*Heterorhabditis indica* @ 15–20 mL / 1 billion infective juveniles per mL).

3. Chemical Control and Stem Injection Protocols

- **Prophylactic Foliar/Stem Sprays:** Spray the pseudostem (especially during the 5th month) with 'Neemosol' @ 12.5 mL/L, Chlorpyrifos 20 EC @ 2.5 mL/L (+ 1 mL/L wetting agent), Azadirachtin (5 mL/L) 2–3 times at 3-weekly intervals, Endosulfan (0.04%), or Carbaryl WP (0.1%). Soil application of Carbofuran granules @ 3 g/stool is also effective.
- **Stem Injection Protocol (For Infestation After 7 Months):**

  - Using a pseudostem injector / injection gun, inject 2 mL of Monocrotophos solution (150 mL Monocrotophos mixed in 350 mL water) or Dichlorvos (0.25%) at 2 feet and 4 feet heights on opposite sides of the stem at a 30-degree downward angle.
  - Alternatively, insert 3 tablets of Celphos per plant into a pseudostem slit and seal the opening with mud plaster to kill all insect stages.

- **Safety Restrictions and Technical Qualifications:**
  - *Timing Limit:* **Strictly no stem injections should be administered after flowering.**
  - *Depth Limit:* The injection needle must penetrate only to a depth of **2 to 3 leaf sheaths**, taking extreme care not to pierce or injure the central growing core.

Sources Cited

1. *Banana Expert System* (URL; Sections: Fifth Month, Sixth Month, Seventh Month).
2. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 12.2: Pseudostem Borer, pp. 20–21).
3. *Banana_pests_3.pdf* (Section: Pseudostem Borer, p. 1).
4. *IPM-Banana-Revised-Sept2011.pdf* (National Horticulture Mission, 2012; Section 2: Pseudo stem borer, pp. 2–3).

1.2 Banana Rhizome Weevil / Corm Weevil (*Cosmopolites sordidus*)

Common and Scientific Names, Taxonomy, and Classification

- **Common Names:** Rhizome Weevil, Corm Weevil, Banana Weevil Borer, Banana Weevil.
- **Scientific Name:** *Cosmopolites sordidus* (Germar).
- **Taxonomic Classification:** Order Coleoptera, Family Curculionidae.

Life Cycle and Developmental Stages

- **Egg:** Elongate-oval in shape, measuring approximately 2 to 3 mm long, and white in color. Eggs are inserted singly into small cavities chewed out by female weevils at the pseudostem base (just above ground level), the upper corm region, shallow roots near the soil surface, cut stumps, or decaying pseudostems. The egg incubation period lasts between 4 and 36 days depending on temperature.
- **Larva (Grub):** The grub represents the primary damaging stage. Grubs are white, fleshy, legless, and possess a distinct reddish-brown head capsule. Larvae bore into and feed extensively within the corm and rhizome tissue, creating winding internal tunnels.

- **Pupa:** Pupation takes place inside the larval feeding tunnels within the corm.
- **Adult:** Free-living, stout weevils that turn dark brown to black a few days after emergence. Adults reside in soil at the base of banana mats, beneath plant debris, or under newly harvested, decaying pseudostems. Adult weevils are nocturnal and can survive up to two years (and endure up to six months without feeding). However, adults are highly sensitive to desiccation and succumb within 48 hours if maintained on a completely dry substrate.

Feeding Behavior, Symptoms, and Direct vs. Secondary Damage

- **Direct Larval Tunneling:** Grubs bore extensively into the corm and rhizome. In severe infestations, larval tunnels extend several feet upward into the lower pseudostem trunk.
- **Early Diagnostic Symptoms:** Reduced plant vigor, sickly appearance, and the development of faint yellow lines along the top leaf lamina.
- **Advanced Foliar and Canopy Symptoms:** Affected plants exhibit stem tapering toward the crown region, marked reduction in leaf size and total leaf count, leaf yellowing, desiccation, and premature leaf death. Bunches display choked-throat emergence, poor finger development, and small hand size.
- **Structural and Vascular Failure:** Internal corm destruction disrupts nutrient and water translocation to the canopy. Extensive corm hollowing weakens the root anchor, predisposing heavy mats to wind lodging and toppling. Yield losses exceed 50% in ratoon crops.
- **Secondary Damage:** Larval tunneling facilitates invasion by secondary rot-causing micro-organisms, converting solid corm tissue into a soft, decayed, blackened mass.

Spread, Host Attractants, and Favorable Conditions

- **Primary Mode of Spread:** Weevils are disseminated locally and across regions primarily through infected planting material (suckers, corms, and rhizome bits harboring eggs or grubs).
- **Host Volatiles and Aggregation:** Adult weevils are attracted to host mats by volatile compounds emitted from injured corm and pseudostem tissues. Males produce an aggregation pheromone that attracts both sexes.

Field Monitoring and Trapping Thresholds

- **Pheromone Traps:** Install Cosmolure pheromone traps @ 5 traps/ha. Shift trap locations monthly.
- **Split Pseudostem Traps:** Monitor adult populations using 30 cm (or 2-foot) longitudinal split banana pseudostem traps placed with cut surfaces facing the ground at a density of 10–15 traps/ha (or 10–15 traps/acre; up to 40 traps/acre in endemic areas).
- **Disc-on-Stump Traps:** Lay 1 to 2-foot cut pieces of old rhizomes or pseudostems on the plantation floor between mats to attract shelter-seeking adults.

Integrated Pest Management (IPM)

1. Cultural Practices and Sanitation

- **Field Sanitation:** Clean fields regularly by removing dead leaves, dried trash, and crop residues. Destroy debris by burning or burying in mulching pits covered with soil.
- **Clean Planting Material and Paring:** Select healthy suckers exclusively from uninfested fields. Pare corms by trimming off all roots and outer necrotic skin layer to expose clean white tissue.
- **Manual Trapping and Collection:** Hand-collect and destroy adult weevils attracted to split pseudostem or disc-on-stump traps.
- **Post-Harvest Management:** Uproot and destroy infested rhizomes and pseudostems immediately after harvest. Avoid *matocking* (leaving harvested pseudostems standing) in weevil-endemic zones.

2. Biological Control Protocols

- **Pre-Planting Corm Dip:** Dip pared corms in a suspension of *Beauveria bassiana* (1×10^9 CFU/mL) for 20 minutes prior to planting to kill eggs and young grubs.
- **Biocontrol Swabbed Traps:** When monitoring traps detect adult weevils, increase trap density to 100 traps/ha. Swab the cut face of longitudinal split stem traps with one of the following biocontrol agents and place cut-side down against the soil:
  - Liquid formulation of *Beauveria bassiana* (1×10^9 spores/mL) or 20 g rice chaffy grain formulation (1×10^8 spores/mg).
  - Entomopathogenic nematode *Heterorhabditis indica* @ 20 mL/trap (or 20 mL at 1 billion infective juveniles per mL).
- **Stump Biocontrol Treatment:** Treat harvested, uprooted pseudostems and cut stumps with *Beauveria bassiana* liquid formulation to eliminate lingering adult populations.

3. Chemical Interventions

- **Pre-Planting Sucker Dips:** Dip suckers before planting in:
  - Monocrotophos 0.5% solution for 30 minutes, or
  - Chlorpyrifos 20 EC solution @ 2.5 mL/L.
- **Prophylactic Soil Granular Application:** Apply Carbofuran 3G @ 20 g/plant around the root zone at the 3rd, 5th, and 7th months after planting.
- **Post-Planting Spray and Base Drenching:** In case of post-planting infestation, spray the pseudostem and drench soil around the mat base with Chlorpyrifos 20 EC @ 2.5 mL/L. Follow one week later with a spray and soil drench of Malathion 50 EC @ 2 mL/L.

Sources Cited

1. *Banana Expert System* (URL; Section: Fifth Month).
2. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 12.1: Rhizome Weevil, pp. 17–19).
3. *Banana_pests_3.pdf* (Section: Rhizome Weevil, p. 2).
4. *IPM-Banana-Revised-Sept2011.pdf* (National Horticulture Mission, 2012; Section 1: Rhizome (corm) weevil, pp. 1–2).

1.3 Banana Aphid (*Pentalonia nigronervosa*)

Common and Scientific Names, Taxonomy, and Classification

- **Common Names:** Banana Aphid, Banana Black Aphid.
- **Scientific Name:** *Pentalonia nigronervosa* Coquerel.
- **Taxonomic Classification:** Order Hemiptera, Family Aphididae.

Life Cycle, Developmental Stages, and Colony Habits

- **Nymphal Stages:** Nymphs are yellowish-green to dark reddish-brown. They pass through four nymphal instars over a development period of 8 to 12 days.
- **Adult Morphology:** Adults are small, dark reddish-brown to black, shiny insects. Both wingless (apterous) and winged (alate) forms occur. Winged adults possess characteristic transparent wings with prominent, heavy dark brown to black venation.
- **Reproduction and Longevity:** Female adults reproduce parthenogenetically, giving birth directly to 35–50 live nymphs over an adult lifespan of 27 to 37 days.
- **Generations and Cycle Duration:** The entire life cycle completes in 12 to 15 days under favorable conditions. Between 30 and 40 overlapping generations are produced annually.
- **Colony Microhabitat:** Aphids congregate in dense colonies hidden inside leaf axils, within tender unfurled leaves, around petiole bases, beneath outer pseudostem sheaths near ground level, and on young emerging suckers.

Seasonal Occurrence, Favorable Conditions, and Host Range

- **Favorable Environmental Conditions:** High atmospheric humidity favors rapid aphid multiplication and colony expansion.
- **Seasonal Dynamics:** Aphids remain present year-round. However, vector activity and population movement peak during winter in plain regions and during summer in hill banana zones.
- **Host Plants:** Banana (*Musa* spp.) is the primary cultivated host. Volunteer banana plants, wild *Musa* species, and adjacent weeds act as alternate hosts during off-seasons.

Feeding Behavior and Direct Damage vs. Secondary Disease Transmission

- **Direct Feeding Damage:** Both nymphs and adults use piercing-sucking mouthparts to extract cell sap from tender foliage, leaf axils, and pseudostems, devitalizing the host plant. Direct feeding leads to local tissue discoloration, malformation, leaf dwarfing, leaf curling, petiole elongation, reduced bunch size, and distorted fingers. Direct feeding injury alone is generally considered minor relative to vector impact.
- **Secondary Damage (Viral Vector Role):** The primary economic threat of *P. nigronervosa* lies in its role as the persistent vector of **Banana Bunchy Top Virus (BBTV)** and a vector of **Banana Bract Mosaic Virus (BBMV)**.
- *Source Disagreement Note on Vector Transmission Mode: Banana_disease_detection.pdf* (p. 36) describes BBTV vector transmission by *P. nigronervosa* as occurring in a **persistent manner**, whereas *IPM-Banana-Revised-Sept2011.pdf* (p. 15) reports transmission as occurring in a **semi-persistent manner**.
- **Vector-Induced Viral Symptoms:** Transmission of BBTV leads to dark green "dot-and-dash" streaking along secondary leaf veins and petioles, severe stunting, narrow upright brittle leaves with rolled margins, a crowded rosette "bunchy top" crown, complete bunch abortion, or production of small, distorted, unmarketable fruits.

Field Monitoring Regularly inspect leaf axils, petiole bases, rolled central leaves, and young suckers for the presence of dark shiny aphid colonies, ant activity (attracted to honeydew), or early viral symptoms on secondary veins.

Integrated Pest Management (IPM)

1. Cultural Practices, Field Sanitation, and Rouging

- **Prerouging Vector Suppression:** Always spray aphid-infested plants with systemic insecticides prior to uprooting or rouging virus-infected mats to prevent winged vectors from dispersing to surrounding healthy plants.
- **Sanitation and Weed Management:** Maintain fields and surrounding borders strictly free of weeds and volunteer plants that serve as off-season aphid reservoirs.
- **Desuckering:** Regularly prune surplus side suckers to eliminate sheltered feeding niches.

2. Conservation of Natural Enemies Protect endemic natural predators that prey on banana aphid colonies:

- **Ladybird Beetles:** *Scymnus nubilus* and *Pseudaspidimerus trinotatus.*
- **Syrphid Fly Larvae:** *Paragus serratus.*
- **Hemerobiid Lacewings:** *Micromus timidus.*

3. Biological and Botanical Interventions

- **Entomopathogenic Fungal Spray:** Apply foliar sprays of *Verticillium lecanii* @ 20 g/L (or 5 mL/L) at 25, 65, and 165 days after planting, directing sprays into leaf axils and pseudostem bases.
- **Neem Cake Application:** Apply neem cake @ 1 kg/plant split into two equal doses at 1 and 2 months after planting.
- **Botanical Sprays:** Spray 2% neem oil garlic emulsion or Neemkosol @ 2.5 mL/L into leaf axils and lower stem bases.
- **Insecticidal Soap:** Thoroughly spray soapy water or insecticidal soap onto petiole bases, furled leaves, whorls, and young suckers.

4. Chemical Insecticide Recommendations and Dosages

- **Foliar Chemical Sprays:** Apply one of the following systemic insecticides upon detecting aphid infestation:
  - Dimethoate 30 EC @ 1 mL/L to 2 mL/L (or 1.485–1.980 L/ha in 1500–2000 L water/ha) at 3 to 4-week intervals.
  - Oxydemeton-methyl / Methyl demeton 25 EC @ 2 mL/L (or 3–4 L/ha / 0.05% a.i.) at 3 to 4-week intervals.
  - Monocrotophos 36 SL @ 0.05% (2 mL/L) or Malathion 50 EC @ 0.1% (2 mL/L) at 10 to 15-day intervals.
- **Soil Granular Application:** To prevent aphid recurrence in endemic areas, apply Phorate 10G @ 1.0 kg a.i./ha, or Carbofuran 3G @ 166 g/plant (or 25 g/pit) around the root zone.
- **Pseudostem / Corm Injection:** Inject 0.05 to 0.2 mL of systemic insecticide (such as Monocrotophos or Dimethoate) per plant into the pseudostem/corm at 3 to 4-week intervals.

Sources Cited

1. *Banana_disease_1.pdf* (Section: Banana Bunchy Top Virus, p. 3).
2. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 12.3: Banana Aphid, pp. 21–22; Section 13.6: Bunchy top, p. 36; Section 18: Approved pesticides, p. 45).
3. *Banana_pests_3.pdf* (Section: Banana Aphid, p. 3).
4. *IPM-Banana-Revised-Sept2011.pdf* (National Horticulture Mission, 2012; Section 3: Banana aphid, p. 4; Section 15: Bunchy Top, pp. 15–17).

1.4 Banana Thrips (*Helionothrips kadaliphilus* and *Thrips hawaiiensis*)

Common and Scientific Names, Taxonomy, and Classification

- **Common Names:** Leaf Thrips, Flower Thrips, Rust Thrips, Bunch Thrips.
- **Scientific Names:**
  - Leaf Thrips: *Helionothrips kadaliphilus*.
  - Flower Thrips: *Thrips hawaiiensis*.
- **Taxonomic Classification:** Order Thysanoptera, Family Thripidae.

Morphology, Life Cycle, and Developmental Stages Thrips undergo complete metamorphosis comprising egg, larva (nymph), pupa, and adult stages:

- **Leaf Thrips (*Helionothrips kadaliphilus*):** Adults possess a reddish-brown body with yellowish-grey wings. They congregate in colonies on the ventral (lower) surface of banana leaves to suck cell sap.
- **Flower Thrips (*Thrips hawaiiensis*):** Adults and nymphs feed on flower tips, floral bracts, and immature fruits before the floral bud emerges from the pseudostem, persisting for up to two weeks after bud emergence.
- **Soil Pupation:** Following active nymphal feeding on foliage and fruit, mature larvae drop to the soil surface to complete pupation.

Feeding Behavior, Symptoms, and Fruit Damage

- **Early Diagnostic Symptoms:** Feeding and oviposition between adjacent or touching green fruits produce water-soaked, smoky, discolored areas where colonies aggregate.
- **Rust Formation and Skin Blemish:** Affected fruit surfaces develop a characteristic rusty-red, reddish-brown, or dark brown-black discoloration (rust spots and linear streaks).
- **Severe Fruit Cracking:** In severe infestations, fruit skin loses elasticity, develops deep longitudinal cracks, and fruits may split open.
- **Foliar Symptoms:** Feeding by leaf thrips causes yellowish-brown feeding spots along the lower leaf surface, leading to leaf chlorosis, desiccation, and premature drying.
- **Varietal Susceptibility:** Cultivars identified as particularly susceptible include Poovan, Nendran, and Karpuravalli.
- **Commercial Impact:** While internal fruit pulp texture and taste remain unaffected, external skin blemish, rust streaks, and cracking render bunches unmarketable for commercial sale and export.

Field Monitoring and Sampling Procedures

- **Sticky Water Traps:** Install blue pan water sticky traps @ 4–5 traps/acre to monitor adult thrips activity.
- **Visual and Tapping Counts:** Monitor thrips populations by counting nymphs and adults on 5 terminal leaves per plant (using visual inspection or the tapping method).
- **Bunch Inspection:** Conduct regular physical checks of developing fruit beneath bunch covers to detect early feeding water-soaked spots before rust streaks expand.

Integrated Pest Management (IPM)

1. **Cultural, Physical, and Sanitation Practices**

- **Clean Planting Material:** Plant exclusively thrips-free suckers or tissue-cultured banana plantlets.
- **Hot Water Sucker Treatment:** Submerge pared suckers in hot water (50–55°C) prior to field planting.
- **Eradication of Host Residues:** Destroy all volunteer banana plants, weeds, and alternate host vegetation in and around the plantation that act as off-season thrips reservoirs.
- **Bunch Sleeving / Bagging:** Sleeve developing bunches at shooting using polypropylene covers (covering the full length of the bunch) immediately after removal of the male flower bud to physically protect fingers from thrips invasion.

2. **Biological Control**

- **Soil Biopesticide Drenching:** Apply a liquid formulation of the entomopathogenic fungus *Beauveria bassiana* (1×10^9 spores/mL) to the soil around the mat base to target and kill pupating thrips in the soil.

Sources Cited

1. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 7: Sampling procedure for thrips, p. 16; Section 12.5: Thrips, pp. 23–25; Section 19: References, p. 46).

1.5 Plant-Parasitic Nematode Complex (Burrowing, Root-Knot, Root-Lesion, and Spiral Nematodes)

Common and Scientific Names, Classification, and Feeding Guilds Plant-parasitic nematodes represent a major subterranean threat to banana production. The primary species identified across available sources are categorized by feeding habits and root injury morphology:

- **Lesion-Forming Endoparasites:**

  - *Burrowing Nematode: Radopholus similis* (Cobb) Thorne (Migratory endoparasite).
  - *Root-Lesion Nematode: Pratylenchus coffeae* (Zimmermann) Filipjev & Schuurmans Stekhoven (Migratory endoparasite).
  - *Spiral Nematode: Helicotylenchus multicinctus* (Cobb) Golden (Ecto- and endoparasite).

- **Gall-Forming Endoparasites:**
  - *Root-Knot Nematodes: Meloidogyne incognita* (Kofoid & White) Chitwood and *Meloidogyne javanica* (Treub) Chitwood (Sedentary endoparasites).

Life Cycle and Pest Biology

- ***Radopholus similis*:** Female nematodes live 2 to 3 months and lay over 100 eggs inside corm and root tissues. Hatched juveniles and adults migrate continuously through root cortical cells. The complete life cycle requires approximately 2 weeks. Nematodes survive saprophytically or in quiescent states in soil in the absence of a host.
- ***Pratylenchus coffeae*:** Complete development takes less than 20 days when soil temperatures range between 25°C and 30°C.
- ***Meloidogyne* spp.:** Second-stage infective juveniles (J2) penetrate young feeder roots, establish permanent feeding giant cells inside, become sedentary, and complete their life cycle inside root tissue, inducing root swellings.
- ***Helicotylenchus multicinctus*:** Robust nematodes possessing strong stylets that feed ecto- or endoparasitically on outer cortical root cells, impairing feeder root formation.

Feeding Behavior, Symptoms, and Direct vs. Secondary Damage

- **Direct Root and Corm Lesions:** Lesion-forming species (*R. similis*, *P. coffeae*) penetrate and tunnel through root and rhizome cortex tissues, producing characteristic elongate reddish-brown to dark black cortical lesions that are clearly visible when infected roots are split longitudinally. In severe attacks, the root cortex is destroyed, turned black, and girdled.
- **Direct Root Galling:** *Meloidogyne* species induce distinct galls and swellings on feeder roots, severely disrupting water and mineral absorption.
- **Above-Ground Foliar Symptoms:** Affected plants display pale yellow leaves with dried margins (frequently misdiagnosed as nutrient deficiency), reduced plant vigor, stunting, delayed development, choked-throat bunch emergence, and poor finger count and hand weight.
- **Structural Failure ("Toppling Over" / "Tip Over" / "Black Head"):** Destruction of primary cord roots and girdling of anchoring roots undermine structural anchorage. Heavy mats collapse or "topple over" under wind pressure or during fruit bearing. Infested plants fail to respond to fertilizers or irrigation, and damage accumulates severely in ratoon crops.

- **Secondary Pathogen Invasion:** Nematode feeding wounds serve as primary entry points for secondary soil-borne fungal pathogens (such as *Fusarium* spp.), accelerating root decay and vascular collapse.

Inoculum Sources, Survival, and Spread

- **Primary Source of Inoculum:** Nematode-infested soil and symptomless, infected planting suckers or corm bits.
- **Secondary Spread:** Free-living second-stage juveniles (J2) migrating through soil water films and irrigation runoff.

Integrated Nematode Management (INM)

1. Cultural, Physical, and Sanitation Practices

- **Clean Planting Material and Corm Paring:** Select suckers exclusively from uninfested land. Pare/trim corms by removing all roots and outer necrotic skin until clean white tissue is exposed.
- **Hot Water Sucker Treatment:** Submerge pared suckers in a hot water bath at 50–55°C for 15 to 30 minutes (or 20 minutes) to eliminate internal nematodes prior to planting.
- **Soil Solarization and Fallowing:** Cover tilled, irrigated soil with plastic mulch for 6 to 8 weeks. Fallowing land for 3 months after harvest suppresses *R. similis*. Flood fallowing for 5 months destroys both *R. similis* and *Fusarium* species.
- **Crop Rotation and Antagonistic Intercropping:** Rotate infested fields with non-host crops such as wetland paddy, sugarcane, cotton, green gram, sunn hemp, tapioca, or turmeric. Intercrop banana with marigold (*Tagetes* spp.) or sunn hemp (*Crotalaria juncea*) as trap crops. Avoid excessive nitrogen applications.
- **Organic Soil Amendments:** Apply 5–10 kg FYM per plant. Apply neem cake @ 400 g to 500 g per plant at planting and at 4 months after planting (or 250–400 g/pit). Incorporate oil cakes of neem, mahua, castor, or karanji.
- **Sludge/Compost Mixture:** Apply distillery sludge (2.5 kg) + vermicompost (1 kg) + neem cake (1 kg) + poultry manure (2.5 kg) per plant at the 3rd, 5th, and 7th months after planting.

2. Biological Control Protocols

- ***Trichoderma* Bio-Enrichment:** Apply *Trichoderma harzianum* 1.5% WP (Strain IIHR-TV-5 / ITCC 6889) or *Trichoderma viride* enriched FYM @ 2 kg/plant at planting and at 3-month intervals.
- **Sucker Dip Treatment:** Dip pared corms in a suspension of *Beauveria bassiana* (1×10^9 CFU/mL) for 20 minutes, or in neem formulations (Nimbecidine, Juerken, or Neewin @ 15 mL/L water) for 30 minutes.

- **Biocontrol Organisms and Botanicals:** Soil application of *Paecilomyces lilacinus*, *Pasteuria penetrans*, or mycorrhizae (*Glomus fasciculatum / Glomus mosseae*) with neem cake. Foliar or soil application of botanical extracts from *Azadirachta indica*, *Calotropis procera*, *Datura stramonium*, *Vitex negundo*, *Solanum torvum*, or *Crotalaria juncea*.

3. Chemical Nematicide Recommendations and CIB&RC Dosages

- **Carbofuran 3G:** Apply @ 20 g to 25 g per pit/plant (or 40 g/plant at planting and at 3-month intervals; or 50 g formulation / 1.5 g a.i. per sucker) at the onset of monsoon rains.
- **Phorate 10G:** Apply @ 10 g per pit at planting.
- **Newer Chemical Nematicides:** Apply Rugby 10G @ 10 g/plant or Caldan 4G @ 10 g/plant at planting and repeat 3 months later.
- **Phenamiphos (Nemacur):** Apply granular Phenamiphos around the mat base (3 applications) for systemic nematode suppression.

Sources Cited

1. *Banana Expert System* (URL; Section: Banana Wilt / Nematode Control).
2. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 12.7: Nematodes, pp. 26–28; Section 12.8: Burrowing nematode, pp. 28–29; Section 18: Approved pesticides, pp. 44–45).
3. *Banana_pests_3.pdf* (Section: Burrowing Nematode, p. 4).
4. *IPM-Banana-Revised-Sept2011.pdf* (National Horticulture Mission, 2012; Section 6: Burrowing nematode, pp. 4–5; Section 7: Root-knot nematode, pp. 5–6; Section 15: IPM/INM Schedule, pp. 17–18).

1.6 Minor Foliar and Fruit Pests (Rugose Spiraling Whitefly, Lacewing Bug, Scarring Beetles, and Leaf Caterpillars)

1. Rugose Spiraling Whitefly (*Aleurodicus rugioperculatus*)

Taxonomy, Classification, and Host Range

- **Scientific Name:** *Aleurodicus rugioperculatus* Martin.
- **Taxonomic Classification:** Order Hemiptera, Suborder Sternorrhyncha, Family Aleyrodidae.
- **Status and Host Range:** Invasive polyphagous pest that severely attacks banana (*Musa* spp.) alongside coconut and palms as highly preferred host plants.

Life Cycle and Developmental Stages

- **Egg:** Short-stalked eggs laid in a characteristic spiraling pattern on the lower leaf surface.
- **Nymph:** Yellowish emerging nymphs that subsequently secrete prominent white waxy filaments.
- **Adult:** Distinctly larger than the common spiralling whitefly, possessing two pale brown wavy markings on the forewings (one medial and one apical).

Feeding Behavior, Direct Damage, and Secondary Sooty Mould

- **Direct Sap-Sucking Damage:** Nymphs and adults feed in dense colonies on the undersides of leaves, extracting host sap and devitalizing foliage.
- **Secondary Sooty Mould Disfigurement:** Feeding whiteflies excrete copious amounts of sweet, watery honeydew that accumulates on upper leaf surfaces. Honeydew deposits attract ants and serve as a substrate for the dark sooty mould fungus, *Capnodium* species. Heavy sooty mould deposition turns leaves black, severely blocking sunlight and impairing photosynthetic efficiency. In severe infestations, complete leaf desiccation and drying occur.

Integrated Management

- **Biological Control and Parasitoid Conservation:** *Encarsia guadeloupae*, an exotic parasitoid introduced in India in 1999, inflicts 60% to 70% natural parasitism on rugose spiraling whitefly populations. Avoid indiscriminate chemical sprays to preserve parasitoids and indigenous natural enemies.
- **Physical Dislodgement and Monitoring:** Install yellow sticky traps to catch flying adults. Spray a 1% starch solution onto affected canopy foliage; as the starch dries and flakes off, it physically dislodges heavy sooty mould crusts from leaf surfaces.
- **Botanical Intervention:** In cases of severe pest buildup, apply foliar sprays of 0.5% neem oil.

2. Banana Lacewing Bug / Tingid Bug (*Stephanitis typicus*)

Taxonomy, Classification, and Morphology

- **Scientific Name:** *Stephanitis typicus* (Distant).
- **Taxonomic Classification:** Order Hemiptera, Family Tingidae.
- **Adult Morphology:** Small, pale bugs possessing transparent, lace-like, glassy wings.

Feeding Symptoms and Leaf Damage Colonies of nymphs and adults feed on the ventral (lower) surface of banana leaves. Sap extraction produces distinct chlorotic and white feeding spots on the upper leaf surface directly opposite feeding sites. The lower leaf surface develops scattered chlorotic spots accompanied by dark brown or black excreta dots.

Integrated Management

- **Physical Monitoring:** Install yellow sticky traps @ 15 traps/ha.
- **Conservation of Natural Enemies:** Protect endemic natural predators, including *Stethoconus praefectus* (Distant) (an obligate predator), as well as egg parasitoids *Erythmelus panis* Enock (reported in Tamil Nadu) and *Anagrus* species (Hymenoptera: Mymaridae; reported in Manipur).
- **Chemical Insecticides and CIB&RC Approved Dosages:** Apply one of the following foliar sprays:
  - Dimethoate 30 EC @ 0.03% a.i. (1.485–1.980 L formulation in 1500–2000 L water/ha).
  - Oxydemeton-methyl 25 EC @ 0.025% a.i. (1.5–2.0 L formulation in 1500–2000 L water/ha).
  - Quinalphos 25 EC @ 0.05% a.i. (3–4 L formulation in 500–1000 L water/ha).

3. Fruit and Leaf Scarring Beetles / Flea Beetles (*Nodostoma subcostatum*, *N. viridipennis*, and *Colaspis hypochlora*)

Scientific Names and Seasonal Occurrence

- **Scientific Names:** *Nodostoma subcostatum* (syn. *Nodocostoma subcostatum*), *Nodostoma viridipennis*, and *Colaspis hypochlora*.
- **Seasonal Incidence:** Population density peaks during the rainy season and humid summer months (April through September), coinciding with flushing periods of new growth.

Feeding Behavior, Symptoms, and Fruit Skin Blemish Adult beetles remain hidden beneath tender unfolded leaves in the central heart whorl and under fruit skin. Beetles feed on young foliage and gnaw/scratch the epicarp of immature green fruits. Feeding injuries produce dark scarred spots, surface pitting, and severe epidermal blemishes. Affected fingers become malformed, stunted, and lose commercial export value.

Integrated Management

- **Cultural Sanitation and Weed Control:** Practice strict orchard hygiene. Remove grass weeds in and around plantations to eliminate breeding harborages and suppress beetle populations without chemical intervention.

- **Chemical Control:** In cases of severe economic loss, apply chemical sprays between April and September (matching new growth emergence):
  - Endosulfan 35 EC @ 0.04%–0.05% (2 mL/L water), or
  - Carbaryl 50 WP @ 0.1% (4 g/L water).
- **Spray Schedule and Safety Precaution:** Repeat sprays at fortnightly intervals if required. **Cease all chemical applications at least 15 days prior to bunch harvest.**

4. Leaf-Eating Caterpillar Complex (*Spodoptera litura* et al.)

Pest Complex Species A complex of lepidopteran larvae feeds on banana foliage, primarily including:

- Tobacco Caterpillar: *Spodoptera litura* (Fabricius).
- Hairy Caterpillars: *Pericallia ricini* (Fabricius), *Asuraila* species, *Creatonotus gangis* Linnaeus, *Diacrisia obliqua* (Walker), *Euproctis fraterna* Moore, and *Eupterote geminata* Walker.
- Additional Lepidopterans: Leaf rollers, bagworms, and slug caterpillars.

Feeding Damage and Symptoms Larvae feed gregariously or individually on leaf laminae, causing extensive defoliation, leaf skeletonization, and shot-hole feeding damage. Leaf roller larvae roll leaf margins into protective tubes tied with silk inside which they feed and pupate.

Integrated Management

- **Mechanical Collection:** Hand-pick and destroy egg masses and early-instar gregarious larvae. Remove and burn rolled leaves harboring leaf roller larvae and pupae.
- **Chemical Insecticides:** During severe rainy season outbreaks, apply foliar sprays of Endosulfan 35 EC @ 2 mL/L or Carbaryl 50 WP @ 4 g/L.

Sources Cited

1. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 12.4: Lacewing Bugs, pp. 22–23; Section 12.6: Rugose Spiraling Whitefly, pp. 25–26; Section 18: Approved pesticides, pp. 45–46).
2. *Banana_pests_3.pdf* (Section: Fruit and Leaf Scarring Beetle, pp. 3–4).
3. *IPM-Banana-Revised-Sept2011.pdf* (National Horticulture Mission, 2012; Section 4: Leaf eating caterpillar, p. 4; Section 5: Flea beetles, p. 4).

1.7 Banana Mealybugs (*Planococcus* spp., *Dysmicoccus brevipes*, et al.)

Species Complex, Taxonomy, and Classification Several mealybug species infest banana plantations, acting as sap-sucking pests and viral vectors:

- *Planococcus citri* (Citrus Mealybug)
- *Planococcus ficus* (Vine Mealybug)
- *Dysmicoccus brevipes* (Pineapple Mealybug)
- *Pseudococcus longispinus* (spelled *Psedococcus longispirus* in *IPM-Banana-Revised-Sept2011.pdf*)
- *Ferrisia virgata* (Striped Mealybug)
- *Saccharicoccus sacchari* (Pink Sugarcane Mealybug)

Taxonomic Classification: Order Hemiptera, Suborder Sternorrhyncha, Family Pseudococcidae.

Feeding Behavior, Microhabitat, and Developmental Coverage

- **Colony Microhabitat:** Mealybugs aggregate in dense, white, cottony colonies around the base of the pseudostem, beneath outer leaf sheaths, and on young suckers near ground level.
- **Direct Sap Extraction:** Nymphs and adults feed on pseudostem sheaths and lower tissue layers, extracting cell sap and causing local tissue weakening.
- **Developmental Stages and Life Cycle:** Specific egg incubation periods, nymphal instar durations, and adult longevity for individual mealybug species are not explicitly detailed in the available sources.

Secondary Viral Vector Role (Banana Streak Virus)

- **Primary Economic Threat:** The primary agricultural threat posed by mealybugs in banana production is their role as vectors of **Banana Streak Virus (BSV)** (Badnavirus).
- **Vector Transmission Species:** *IPM-Banana-Revised-Sept2011.pdf* identifies *Planococcus citri*, *P. ficus*, *Dysmicoccus brevipes*, *Pseudococcus longispinus*, and *Ferrisia virgata* as vectors transmitting BSV from plant to plant. *Banana_disease_detection.pdf* notes that *P. citri* and *Saccharicoccus sacchari* also transmit the pathogen.
- **Non-Mechanical Transmission:** BSV is **not** transmitted mechanically through sap or on tools used during weeding and cultivation; secondary local field spread relies on mealybug vector movement.
- **BSV Symptoms Induced:** Viral transmission leads to golden-yellow chlorotic dots and flecks that elongate into continuous streaks parallel to leaf veins, turning progressively necrotic and black on mature foliage. Severe infections cause leaf stripping (notably in cvs. Poovan, Grand Nain, and Robusta), bunch choking, total bunch abortion, seediness, and distorted fingers with abnormally thin peels.

Transmission, Dissemination, and Inoculum Survival

- **Primary Spread:** Mealybugs and integrated BSV viral sequences spread primarily across regions through infected vegetative planting material (suckers and rhizome bits).
- **Tissue Culture Persistence:** Standard shoot-tip culture does not eliminate mealybug-transmitted BSV from vegetatively multiplied planting material.

Integrated Vector and Pest Management

1. Clean Planting Material and Exclusion

- Establish new plantations using exclusively certified, virus-free suckers or tissue-culture plantlets sourced from disease-free mother blocks.
- Inspect suckers around the pseudostem base and trim/pare outer leaf sheaths prior to planting to ensure the absence of cottony mealybug colonies.

2. Field Sanitation and Eradication

- Regularly monitor pseudostem bases and lower leaf sheaths for mealybug colonies and chlorotic leaf streaking.
- Immediately uproot, remove, and burn or bury infected plants showing BSV symptoms along with their entire rhizome system to prevent mealybugs from acquiring and transmitting the virus to surrounding healthy mats.
- Keep fields and surrounding field borders strictly free of weeds that harbor mealybug populations.

3. Chemical Suppression of Vector Populations

- Suppress mealybug vector populations around the pseudostem base and leaf axils by applying recommended systemic chemical sprays at regular intervals.
- *Note on Chemical Dosages:* While standalone mealybug entries are not listed in the CIB&RC pesticide annexures, systemic sprays recommended for pseudostem/crown sap-sucking insects (such as Dimethoate 30 EC @ 1–2 mL/L or Methyl Demeton 25 EC @ 2 mL/L) are utilized in integrated vector control programs.

Sources Cited

1. *Banana_disease_detection.pdf* (Directorate of Plant Protection Quarantine & Storage, Jan 2022; Section 13.8: Banana streak disease, p. 38).
2. *IPM-Banana-Revised-Sept2011.pdf* (National Horticulture Mission, 2012; Section 14: Banana streak disease, pp. 13–15; Fig. 15).

1.8 Banana Hard Scale Insect (*Aspidiotus destructor*)

Common and Scientific Names, Taxonomy, and Classification

- **Common Name:** Hard Scale Insect.
- **Scientific Name:** *Aspidiotus destructor* Signoret (referenced in source documentation via extension reference *a_destru.htm*).
- **Taxonomic Classification:** Order Hemiptera, Superfamily Coccoidea.

Life Cycle, Biology, and Oviposition

- **Egg Stage and Oviposition Behavior:** Eggs are deposited directly beneath the protective scale covering. As the adult female lays eggs, her body progressively shrinks in size. The female rotates her body as she oviposits, arranging her eggs in distinct concentric circular rings around herself.
- **Egg Batching and Fecundity:** Eggs are laid in small batches of 3 or 4 at a time, with total female fecundity ranging from 65 to 110 eggs.
- **Egg Coloration Changes:** Eggs are white when initially deposited and turn yellow after a few days.
- **Infestation Microhabitat:** Scale colonies infest and encrust the surfaces of banana leaves and pseudostem sheaths.
- **Developmental Timelines:** Specific durations for nymphal instars, male scale pupation, and adult longevity are not explicitly detailed in the available sources.

Feeding Behavior and Direct Damage Hard scale nymphs and adults use sucking mouthparts to extract cell sap from foliage and leaf sheaths. Dense scale encrustation causes leaf chlorosis, yellowing, tissue desiccation, and reduced photosynthetic activity, weakening affected plants.

Integrated Pest Management (IPM)

1. Cultural Practices and Field Sanitation

- **Pruning and Destruction:** Inspect plants regularly and manually collect, prune, and destroy scale-infested leaves and affected plant parts to prevent population spread.

2. Biological Control

- **Coccinellid Predators:** Conserve and utilize predatory beetles to suppress scale colonies. The available sources recommend field releases of the following coccinellid predators:
  - *Chilocorus nigritus*
  - *Scymnus coccivora*

3. Chemical Control

- **Source Qualification:** Specific chemical insecticide recommendations, application rates, or spray schedules for hard scale insects are not provided in the available source annexures.

Sources Cited

1. *Banana.pdf* (Section: AESA based IPM – Banana, p. 17; Section: Hard scale, pp. 28–29).